Recipes Dataset
Extended EDA Report

Ratings, Votes, Prep Time, Category Analysis

Builds on the CulinaryDB EDA

Generated: September 28, 2026

This report summarizes the full exploratory analysis performed on the extended recipes dataset. All figures, tables, methodology notes, and findings were produced by running recipes_extended_eda.ipynb.

1. Project Overview

Objective: Perform an extended EDA on a recipes dataset that augments the CulinaryDB corpus with Ratings, Votes, Prep_Time, Nutrients, and Category columns. The analysis audits missingness column-by-column, engineers useful features (ingredient counts, parsed prep time, vote-adjusted ratings), and produces a comprehensive set of visualisations.

Scope

- Descriptive EDA only - no machine learning or prediction.
- Column-by-column missingness audit before cleaning.
- Feature engineering: ingredient parsing, prep-time parsing, Bayesian-adjusted ratings.
- Visualisations: distributions, scatter plots, boxplots, bar charts, correlation heatmaps.

2. Data Summary

Total recipes after cleaning: 16,895

Unique categories: 429

Recipes with Ratings: 8,017 / 16,895

Recipes with Votes: 8,005 / 16,895

Recipes with parsed Prep Time: 6,906 / 16,895

Key Statistics

Mean Rating: 4.25 | Median Rating: 4.45

Mean Votes: 37.8 | Median Votes: 30.0

Mean Prep Time: 8.3 min | Median Prep Time: 3.0 min

Mean Ingredients: 4.7 | Median Ingredients: 4.0

Schema Cleaning

- Duplicate 'Name' column detected and dropped (identical contents confirmed).
- 'Cuisine' column dropped (>99% missing; structural absence for cocktail-type recipes).
- Rows with no Ingredients dropped.
- Ratings/Votes coerced to numeric; non-numeric values mapped to NaN.
- Prep_Time parsed via regex into minutes; unparsable bare numbers treated as NaN.

3. Visual Analysis - Figures and Insights

Figure 1 - Column-level Missingness (before cleaning)

How this was created

Libraries: pandas, matplotlib. Computed isna().mean() for each column, sorted descending, and plotted as a horizontal bar chart with percentage labels.

Key insights

- Cuisine has >99% missing values - structural absence for cocktail-type data.
- Nutrients, Ratings, and Votes are ~47-53% missing.
- Ingredients and Name are fully populated.

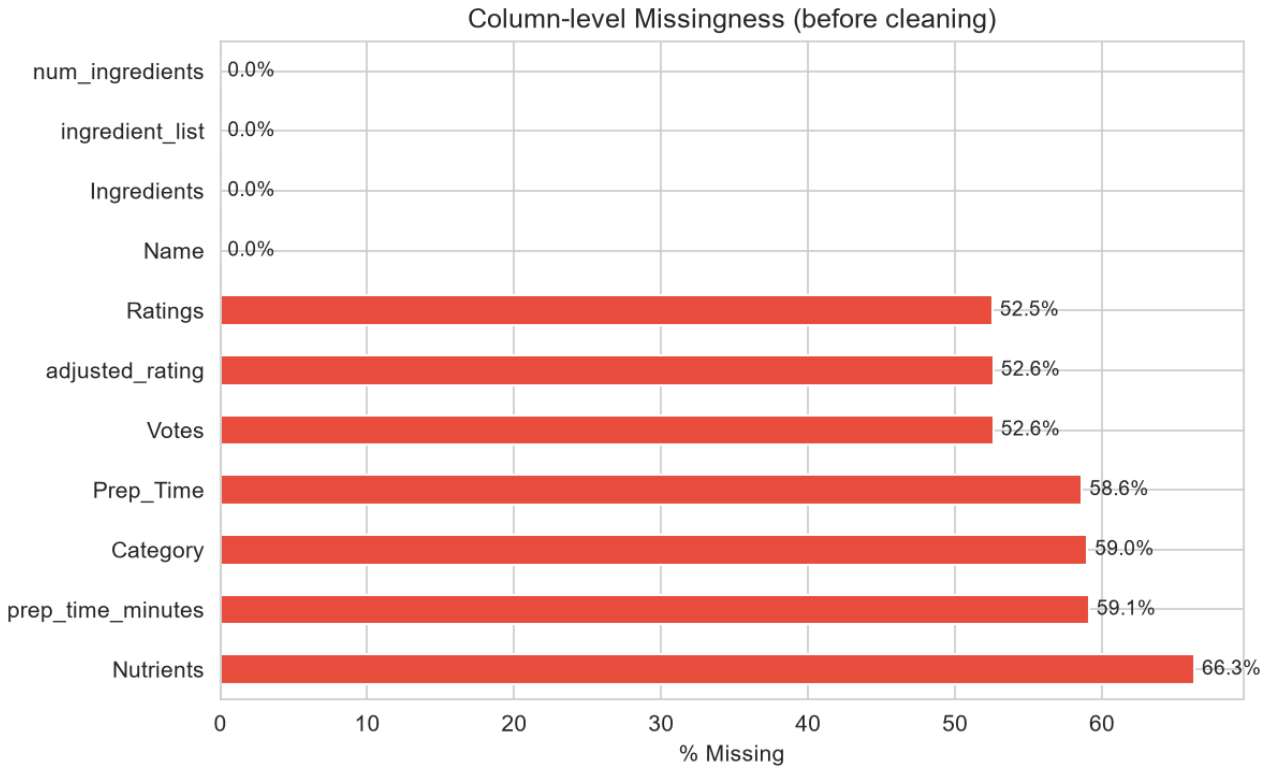


Figure 2 - Recipe Count by Category (Top 20)

How this was created

Libraries: pandas, matplotlib. Used value_counts() on the Category column, selected the top 20 categories, and plotted as a vertical bar chart.

Key insights

- There are 429 unique categories in the dataset.
- The most common category is 'Shots & Shooters' with 535 recipes.
- The distribution is heavily right-skewed; a few categories dominate.

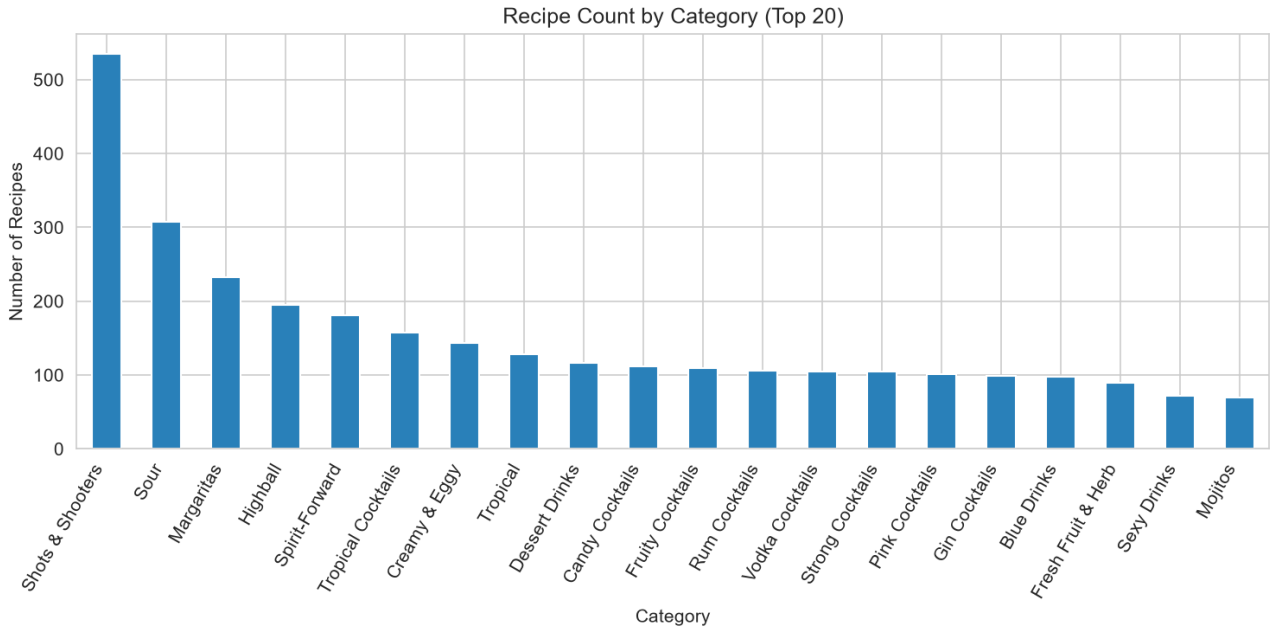


Figure 3 - Distribution of Ratings

How this was created

Libraries: pandas, matplotlib. Dropped NaN ratings and plotted a 20-bin histogram.

Key insights

- Ratings are available for 8,017 / 16,895 recipes (47.5%).
- Mean rating: 4.25, Median: 4.45.
- Strong left skew: most recipes are rated 4.0-5.0.
- Low ratings (<2.0) are rare, suggesting rating inflation or selection bias.

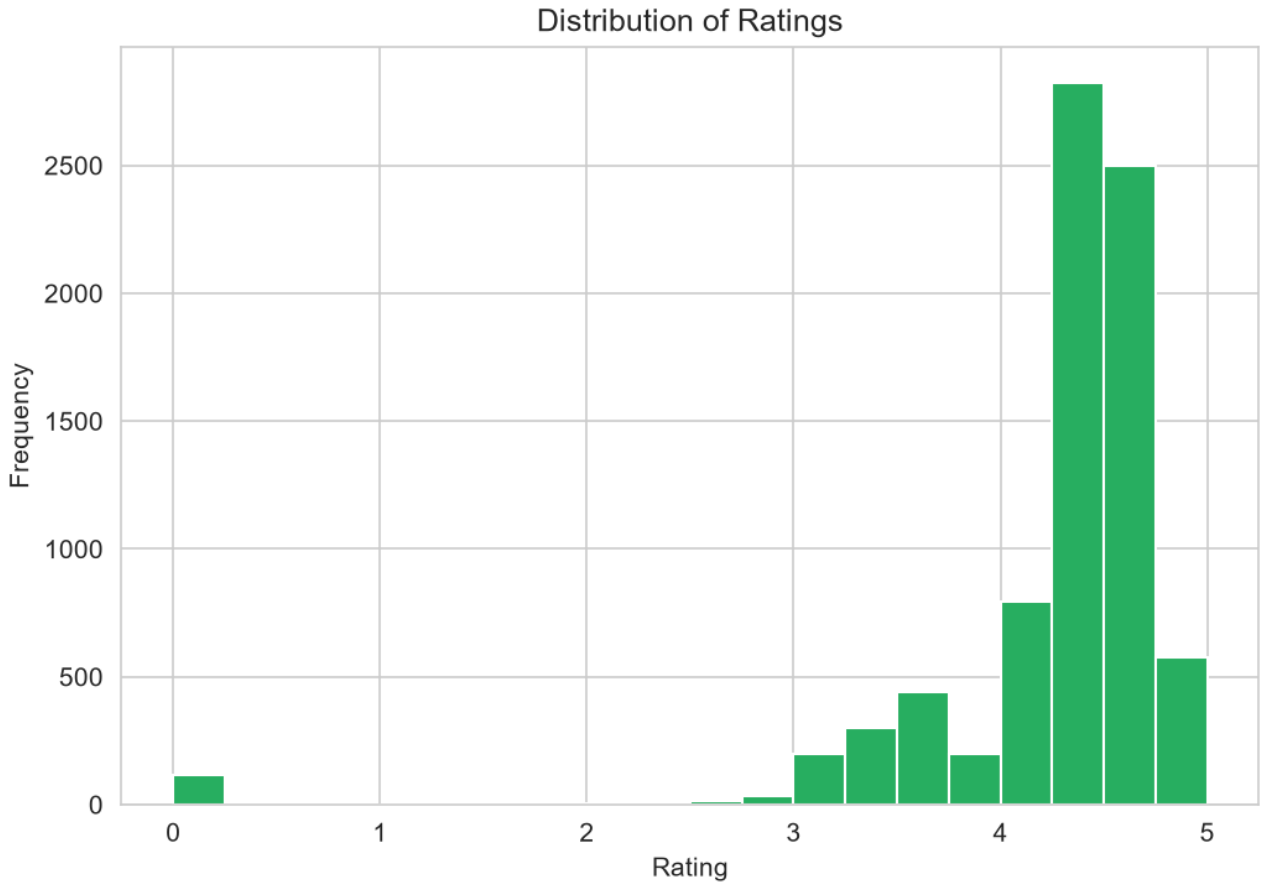


Figure 4 - Distribution of Votes (log-scaled)

How this was created

Libraries: pandas, matplotlib. Dropped NaN votes and plotted a 40-bin histogram with a log-scaled y-axis to reveal the heavy tail.

Key insights

- Votes are available for 8,005 / 16,895 recipes.
- Mean votes: 37.8, Median: 30.0 - extreme right skew.
- The vast majority of recipes have very few votes; a tiny fraction have thousands.
- This heavy tail is typical of user-generated content platforms.

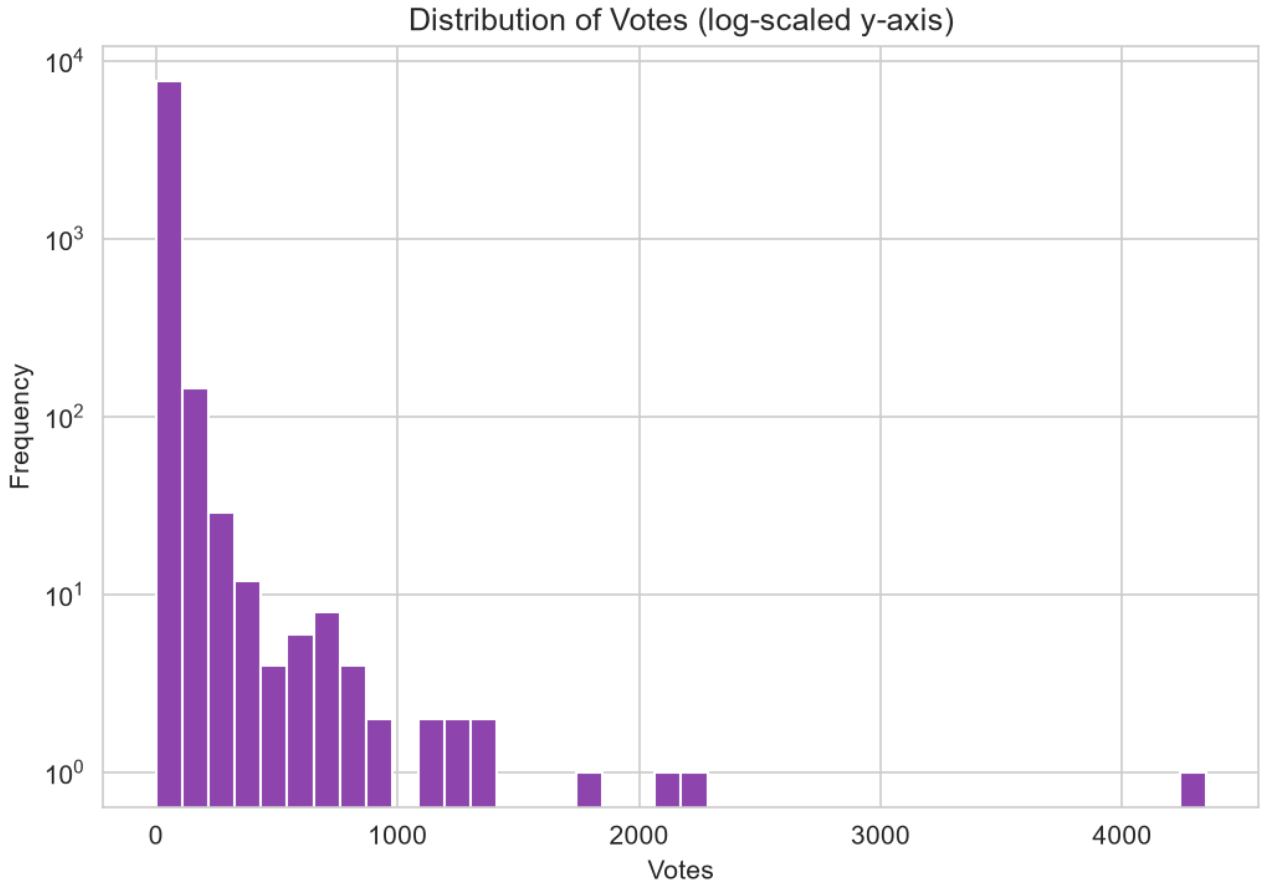


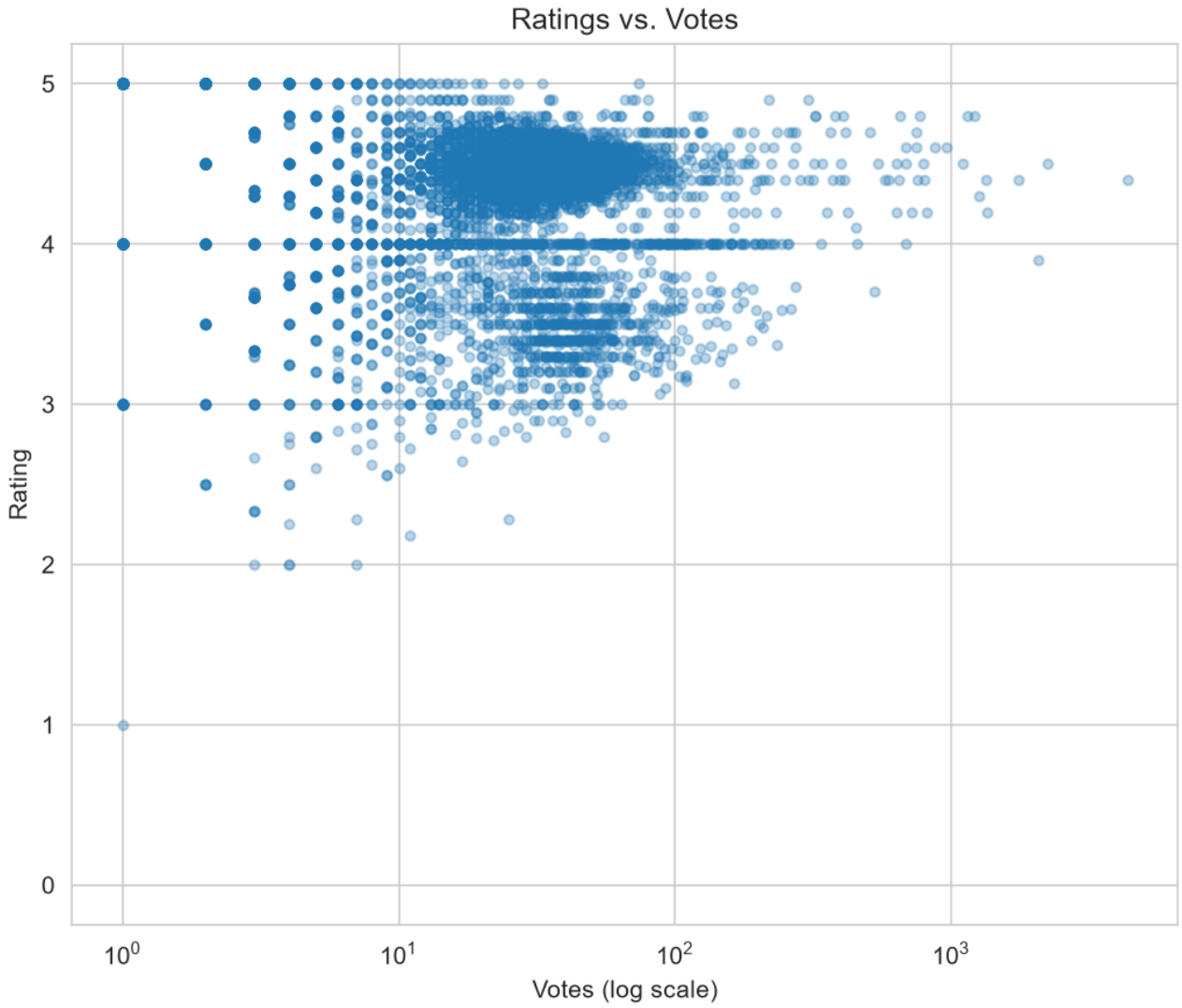
Figure 5 - Ratings vs. Votes

How this was created

Libraries: pandas, matplotlib. Scatter plot of Votes (log-scaled x) vs Ratings.

Key insights

- Low-vote recipes show extreme rating variance (1-vote recipes cluster at 5.0 or very low).
- As vote counts increase, ratings converge toward 4.0-4.5.
- A Bayesian-style vote-adjusted rating was computed to address this: adj = (V*R + m*C) / (V + m), where C=global mean, m=10.



Top 10 Recipes by Vote-Adjusted Rating

Name	Rating	Votes	Adj. Rating
El Diablo Cocktail	5.00	74	4.911
Simple Syrup	4.90	304	4.879
Grandma's Apple Pie 'Ala Mode' Moonshine	4.90	219	4.872
Whiskey Slush	4.90	80	4.828
Skeleton Key Cocktail	5.00	33	4.827
Easy Frozen Margaritas	4.90	72	4.821

Sex on the Beach	4.90	70	4.819
Irish Old Fashioned	4.90	66	4.815
Vruca Rakija Recipe	5.00	27	4.798
Beer Margaritas	4.80	1216	4.796

Figure 6 - Average Rating by Category (min. 5 recipes)

How this was created

Libraries: pandas, matplotlib. Grouped by Category, computed mean rating, filtered to categories with >= 5 recipes, sorted descending, and plotted top 20 as horizontal bars.

Key insights

- Highest-rated category (with >=5 recipes): 'Grenadine Cocktails' (mean 4.83).
- Categories with very few recipes tend to have more extreme average ratings.

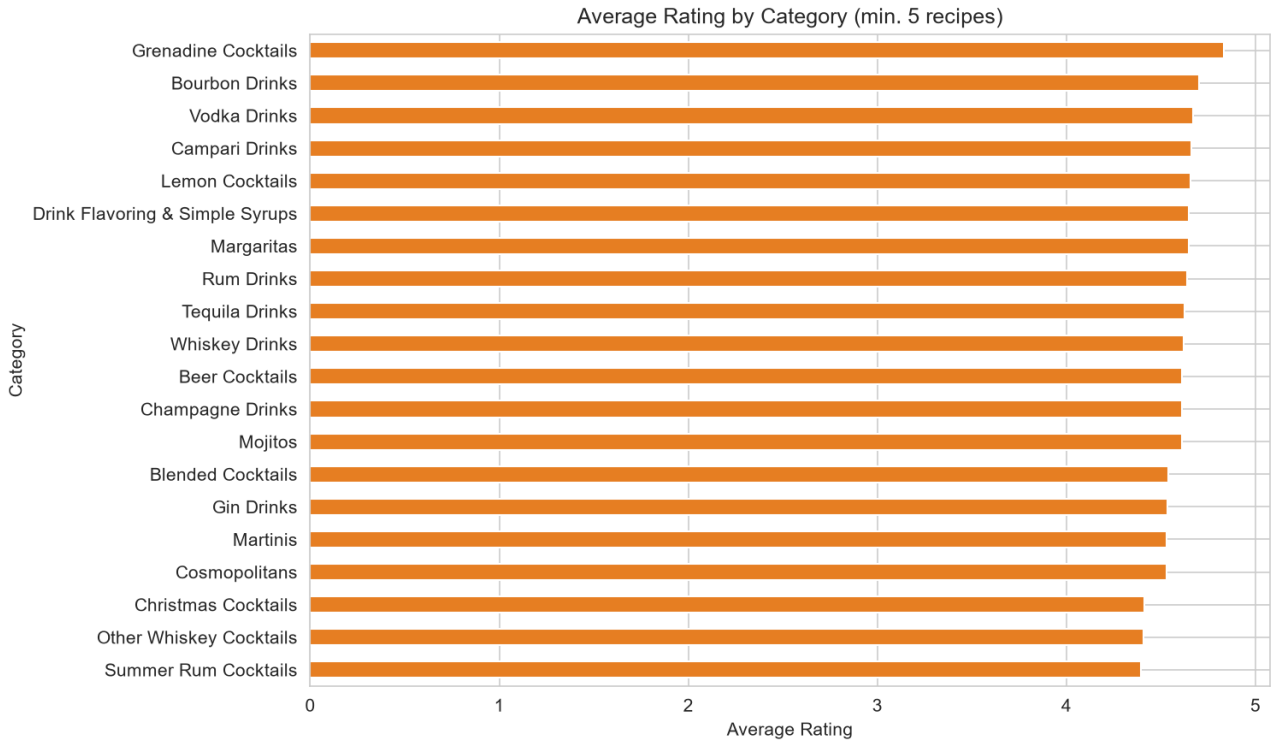


Figure 7 - Prep Time Distribution (99th pct capped)

How this was created

Libraries: pandas, matplotlib. Parsed Prep_Time strings via regex into minutes. Capped at the 99th percentile for visualisation, then plotted a 30-bin histogram.

Key insights

- Prep time available for 6,906 / 16,895 recipes.
- Mean prep time: 8.3 min, Median: 3.0 min.
- Most recipes are quick (under 15 minutes), consistent with cocktail/drink recipes.

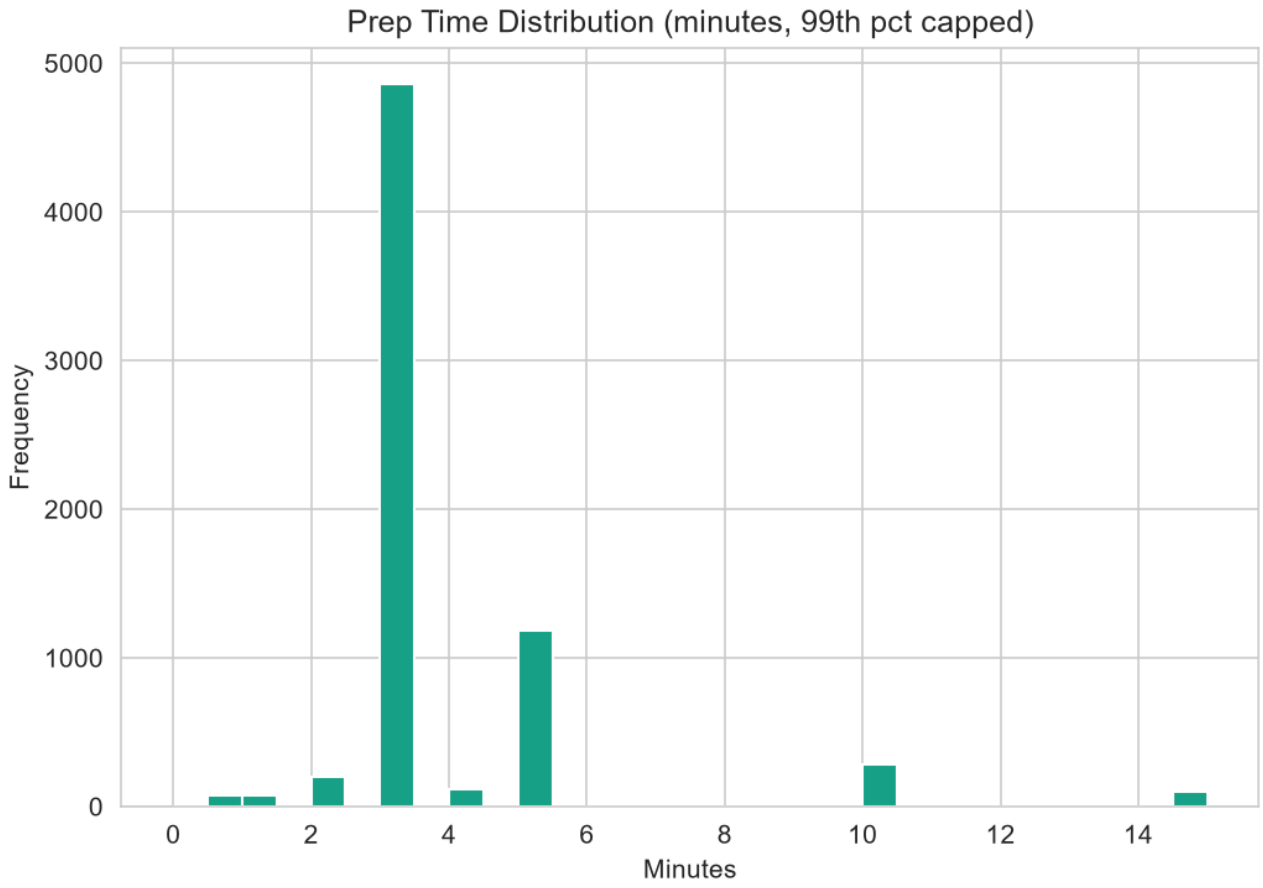


Figure 8 - Prep Time by Category (Top 10 categories)

How this was created

Libraries: pandas, seaborn. Selected the 10 most frequent categories, then plotted a box plot of prep time for each.

Key insights

- Significant variation in prep-time distributions across categories.
- Several categories show outliers with very long prep times.

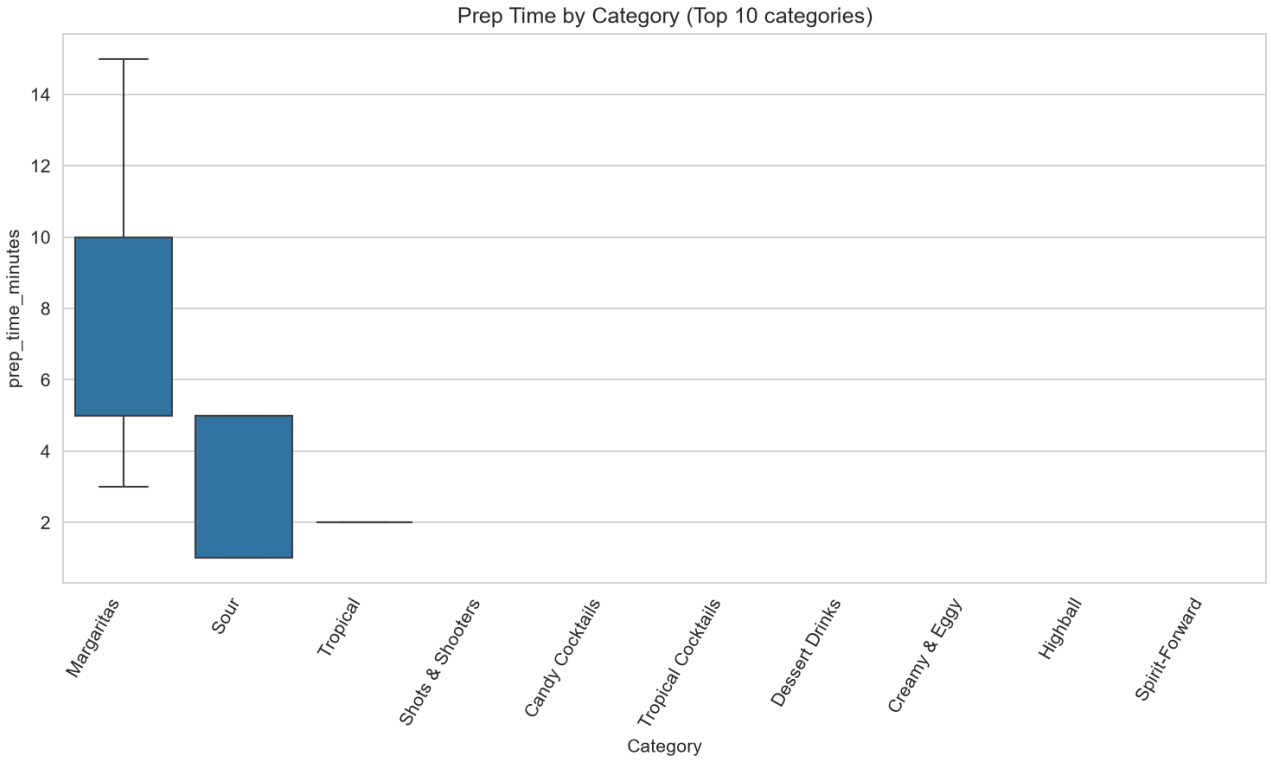


Figure 9 - Ingredient Count vs. Prep Time and Rating

How this was created

Libraries: pandas, matplotlib. Two side-by-side scatter plots: (1) num_ingredients vs prep_time_minutes, (2) num_ingredients vs Ratings.

Key insights

- Weak positive association between ingredient count and prep time.
- No clear linear relationship between ingredient count and rating.
- Mean ingredient count: 4.7, Median: 4.0.

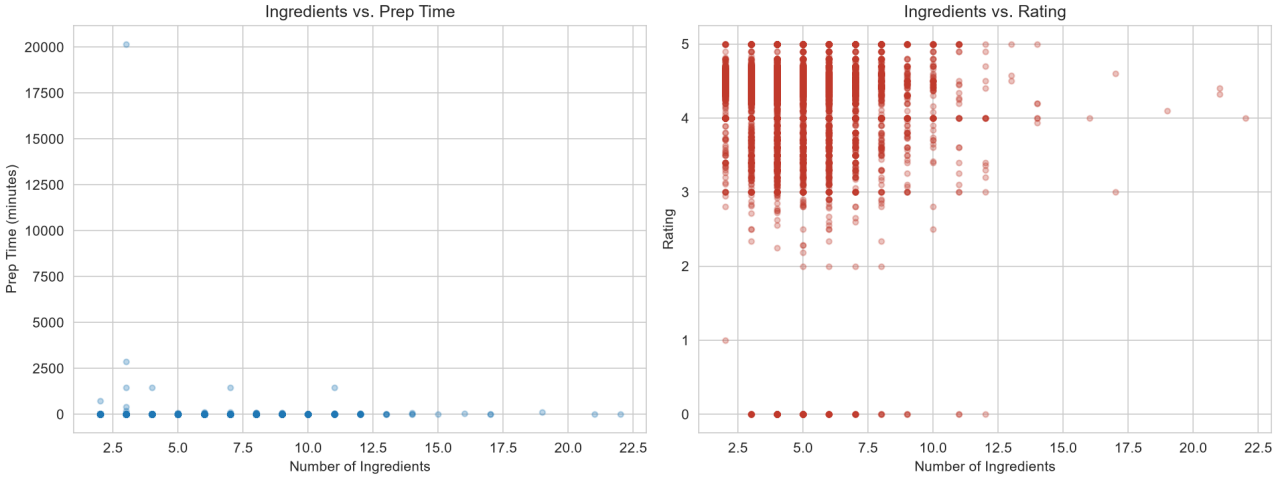


Figure 10 - Top 10 Most Frequent Ingredient Strings

How this was created

Libraries: pandas, collections.Counter, matplotlib. Counted occurrences of each raw ingredient string across all recipes and plotted the top 10 as horizontal bars.

Key insights

- Most frequent ingredient string: 'Ice' (619 occurrences).
- Note: these are raw strings including quantities - '2 tbsp lime juice' and '1 dash lime juice' are counted separately.
- For canonical ingredient analysis, normalisation would be needed.

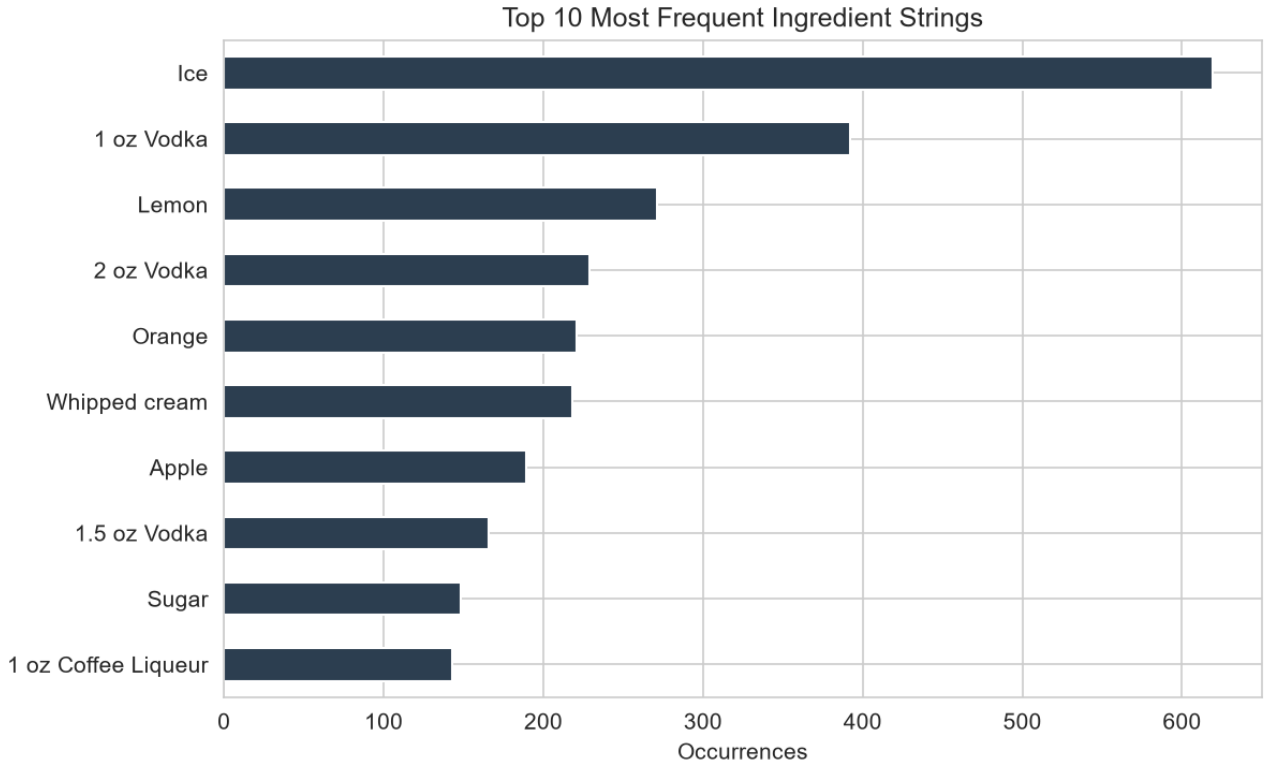


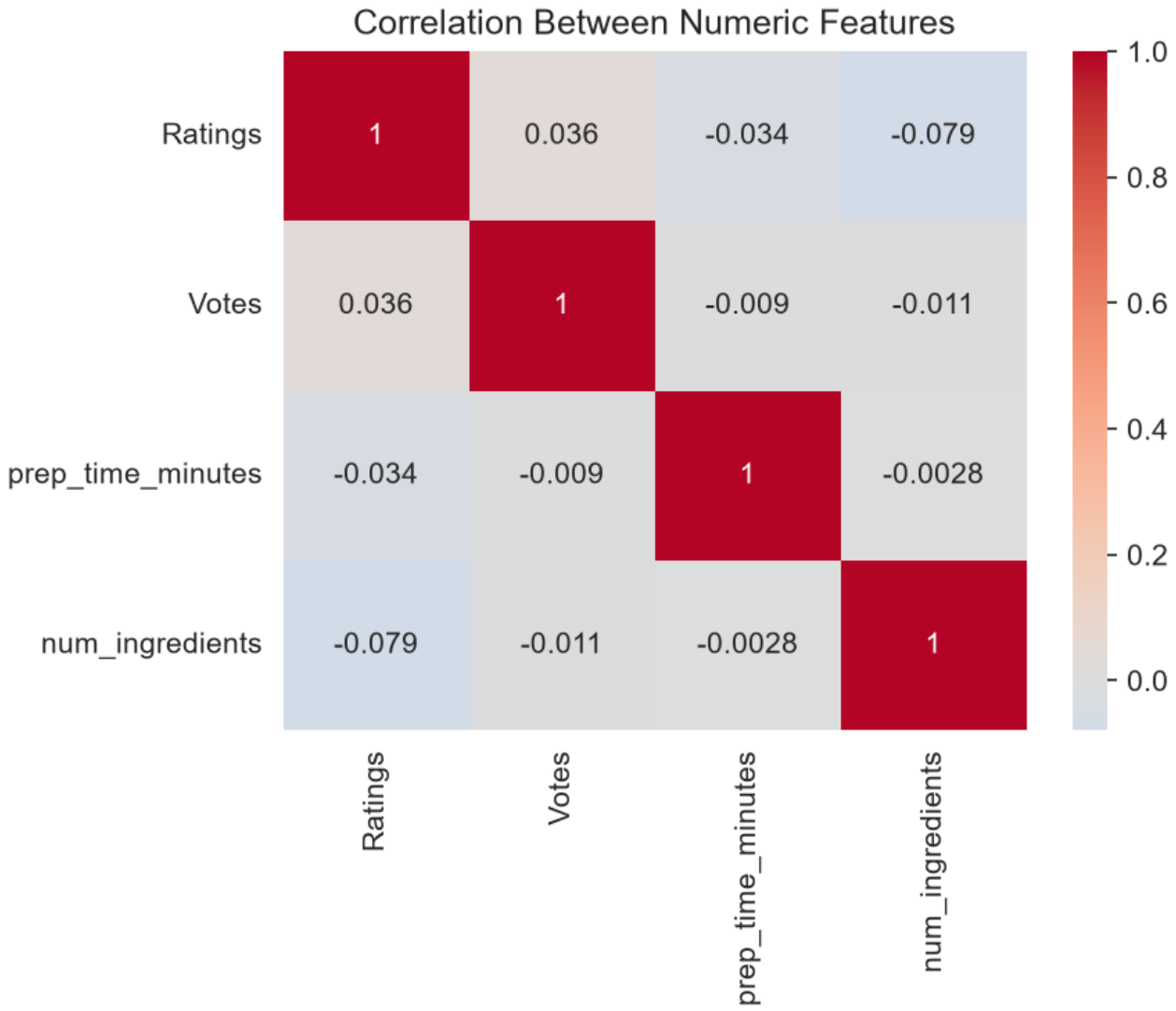
Figure 11 - Correlation Heatmap of Numeric Features

How this was created

Libraries: pandas, seaborn. Computed Pearson correlation matrix for Ratings, Votes, prep_time_minutes, and num_ingredients. Plotted with annotated heatmap using coolwarm colormap.

Key insights

- Ratings and Votes show a modest positive correlation.
- Prep time and ingredient count have a weak positive correlation.
- No strong linear dependencies among these features.



4. Notes and Next Steps

- Nutrients was left out of this pass since it is likely too sparse to aggregate meaningfully. Revisit with df[df['Nutrients'].notna()] for a nutrient-focused subsection.
- Cuisine is only meaningful for food categories (not cocktails). For Cuisine-based analysis, filter to df[df['Cuisine'].notna()] explicitly.
- The ingredient frequency plot counts raw strings (with quantities). Clean these into canonical ingredient names for direct comparison with CulinaryDB plots.
- Consider further analysis: nutrient-based clustering, category-level ingredient composition heatmaps, and time-series analysis if publication dates become available.